Mariano Marcos State University
SDG 15 · Life on Land

STORYBOARD · CHAPTER 11 OF 13

11

Protecting water on university land

3 frames

15 LIFE ON LAND

THE STORY

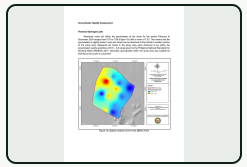
A duty to protect university waters, backed by a monitored aquifer on the MMSU Farm.

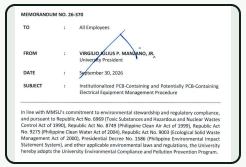
MMSU'S COMMITMENT

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

THE FILMSTRIP · 3 FRAMES

2025 Ongoing programme


1



2

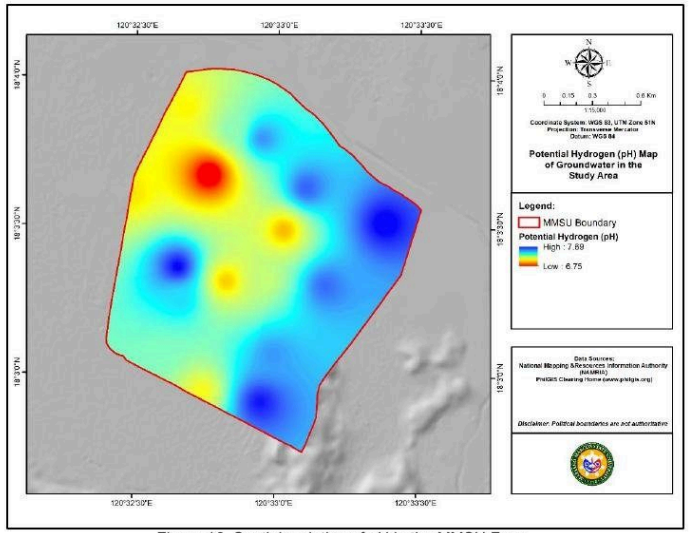

3

AT A GLANCE

17
groundwater wells

6.75–7.69
mean pH, 2024

RA 9275
Clean Water Act



Groundwater Quality Assessment

Potential Hydrogen (pH)

Measured mean pH within the groundwater of the study for the period February to December 2024 ranges from 6.75 to 7.69 (Figure 16) with a mean of 7.33. This means that the groundwater is slightly acidic. Lower pH values can be observed at the central to western portion of the study area. Measured pH levels in the study area were observed to be within the groundwater quality guidelines of 6.5 – 8.5 range given by the Philippine National Standards for Drinking Water (PNSDW) 2017. Generally, groundwater within the study area was suitable for drinking as far as pH is concerned.

Figure 16. Spatial variation of pH in the MMSU Farm

Document: Groundwater Quality Assessment of the MMSU Farm, spatial variation of pH (monitoring through June 2025).

FEB 2024 – JUN 2025 · MMSU FARM

Groundwater Quality Monitoring at the MMSU Farm

1 THE MONITORING

Through June 2025, monthly monitoring of 17 groundwater wells on the MMSU Farm for pH, total dissolved solids, electrical conductivity, chloride and nitrate-nitrogen, compared with the Philippine National Standards for Drinking Water (2017).

2 THE RESULT

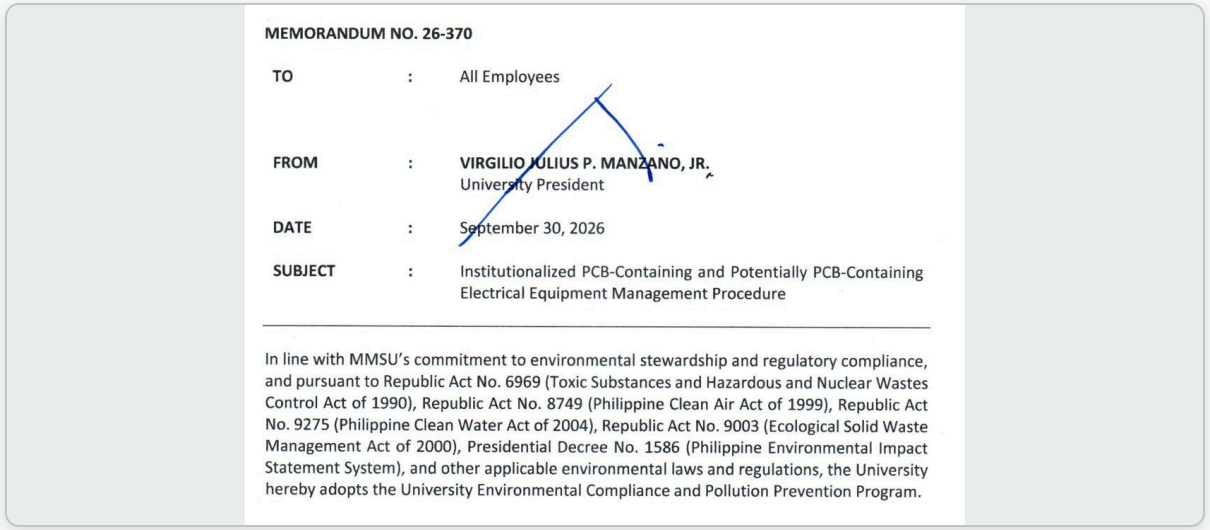
Mean pH from Feb to Dec 2024 ranged from 6.75 to 7.69, within the 6.5–8.5 guideline. Wells with persistently high dissolved solids, which rose again Jan–May 2025, are flagged for continued monitoring.

3 THE USE

Findings guide water resource management and prevent further groundwater degradation; small campus reservoirs also collect rainwater, reduce flooding and recharge groundwater.

17	5	6.75–7.69
monitoring wells	parameters	mean pH (guideline 6.5–8.5)

Source: MMSU, Groundwater Quality Assessment of the MMSU Farm (monitoring February 2024 – June 2025)



MEMORANDUM NO. 26-370

TO : All Employees

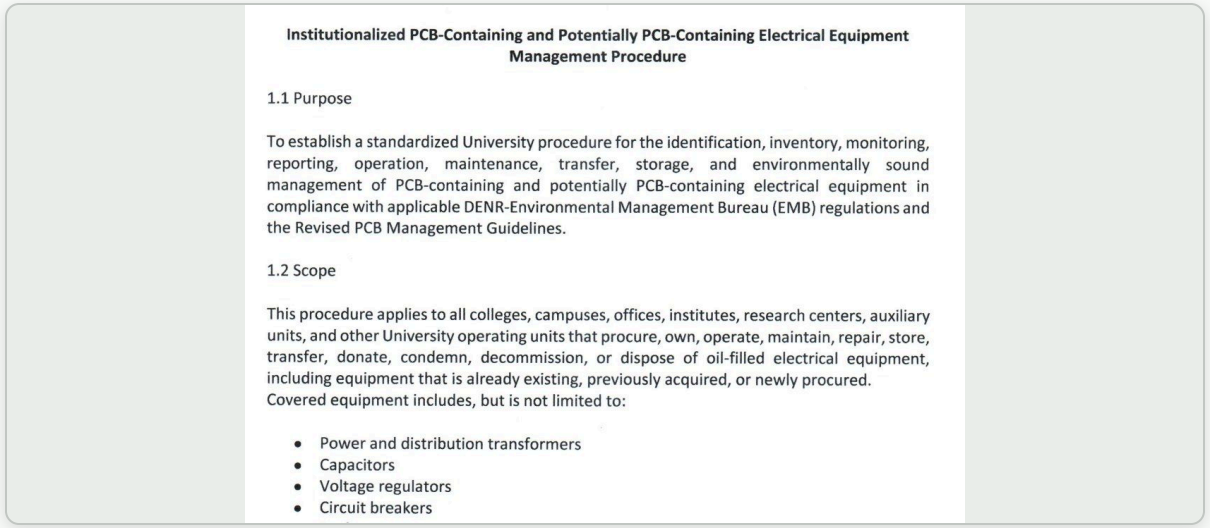
FROM : VIRGILIO JULIUS P. MANZANO, JR.
University President

DATE : September 30, 2026

SUBJECT : Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure

In line with MMSU's commitment to environmental stewardship and regulatory compliance, and pursuant to Republic Act No. 6969 (Toxic Substances and Hazardous and Nuclear Wastes Control Act of 1990), Republic Act No. 8749 (Philippine Clean Air Act of 1999), Republic Act No. 9275 (Philippine Clean Water Act of 2004), Republic Act No. 9003 (Ecological Solid Waste Management Act of 2000), Presidential Decree No. 1586 (Philippine Environmental Impact Statement System), and other applicable environmental laws and regulations, the University hereby adopts the University Environmental Compliance and Pollution Prevention Program.

Document: Office of the President Memorandum No. 26-370, 30 September 2026, adopting the University Environmental Compliance and Pollution Prevention Program.



Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure

1.1 Purpose

To establish a standardized University procedure for the identification, inventory, monitoring, reporting, operation, maintenance, transfer, storage, and environmentally sound management of PCB-containing and potentially PCB-containing electrical equipment in compliance with applicable DENR-Environmental Management Bureau (EMB) regulations and the Revised PCB Management Guidelines.

1.2 Scope

This procedure applies to all colleges, campuses, offices, institutes, research centers, auxiliary units, and other University operating units that procure, own, operate, maintain, repair, store, transfer, donate, condemn, decommission, or dispose of oil-filled electrical equipment, including equipment that is already existing, previously acquired, or newly procured.
Covered equipment includes, but is not limited to:

- Power and distribution transformers
- Capacitors
- Voltage regulators
- Circuit breakers

Document: attached procedure for PCB-containing and potentially PCB-containing electrical equipment (purpose, scope and inventory), Memorandum No. 26-370.

30 SEP 2026 · OFFICE OF THE PRESIDENT MEMORANDUM NO. 26-370

Memorandum No. 26-370 Institutionalizes MMSU's Environmental Compliance and Hazardous Waste Procedures

Ongoing programme · 30 September 2026

1 **THE MEMORANDUM**
President Manzano adopted the University Environmental Compliance and Pollution Prevention Program under RA 6969, 8749, 9275, 9003 and PD 1586, directing all offices, colleges, laboratories, research units and auxiliary enterprises to comply.

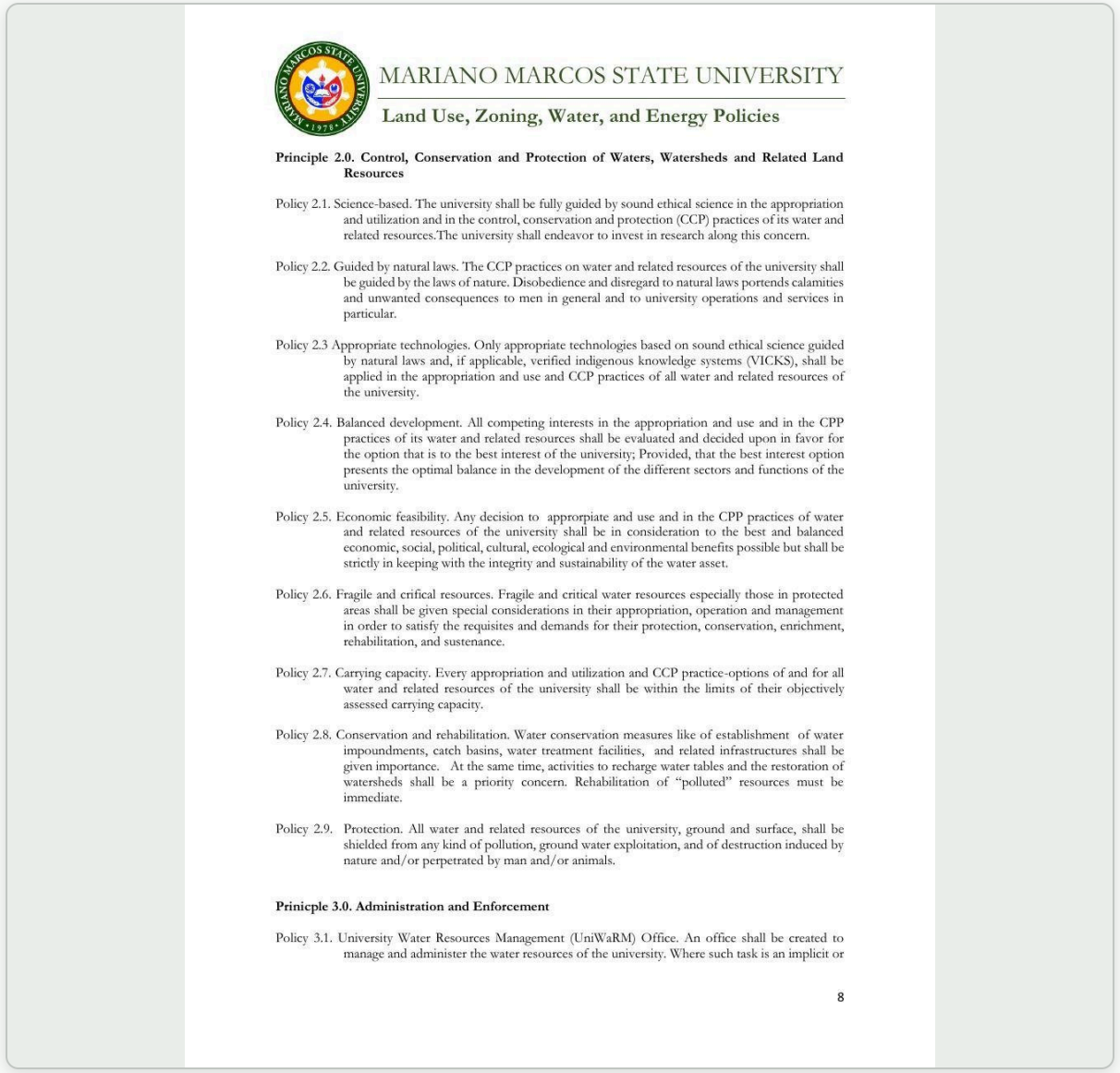
2 **WHAT IT COVERS**
Hazardous wastes, wastewater, air emissions, solid wastes and chemicals, with the permits, monitoring, reporting and inspections required by DENR-EMB.

3 **THE PCB PROCEDURE**
Every unit inventories PCB-containing equipment for the SFT Office, which validates it, keeps the central and DENR-EMB online inventories and inspects annually. Nothing is moved, repaired, condemned or disposed of without SFT written clearance; disposal only through DENR-accredited facilities.

5
laws and decrees cited

30 Sep 2026
issued

Source: MMSU Office of the President, Memorandum No. 26-370, "Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure," 30 September 2026



MARIANO MARCOS STATE UNIVERSITY

Land Use, Zoning, Water, and Energy Policies

Principle 2.0. Control, Conservation and Protection of Waters, Watersheds and Related Land Resources

Policy 2.1. Science-based. The university shall be fully guided by sound ethical science in the appropriation and utilization and in the control, conservation and protection (CCP) practices of its water and related resources.The university shall endeavor to invest in research along this concern.

Policy 2.2. Guided by natural laws. The CCP practices on water and related resources of the university shall be guided by the laws of nature. Disobedience and disregard to natural laws portends calamities and unwanted consequences to men in general and to university operations and services in particular.

Policy 2.3 Appropriate technologies. Only appropriate technologies based on sound ethical science guided by natural laws and, if applicable, verified indigenous knowledge systems (VICKS), shall be applied in the appropriation and use and CCP practices of all water and related resources of the university.

Policy 2.4. Balanced development. All competing interests in the appropriation and use and in the CPP practices of its water and related resources shall be evaluated and decided upon in favor for the option that is to the best interest of the university; Provided, that the best interest option presents the optimal balance in the development of the different sectors and functions of the university.

Policy 2.5. Economic feasibility. Any decision to approrpiate and use and in the CPP practices of water and related resources of the university shall be in consideration to the best and balanced economic, social, political, cultural, ecological and environmental benefits possible but shall be strictly in keeping with the integrity and sustainability of the water asset.

Policy 2.6. Fragile and crifical resources. Fragile and critical water resources especially those in protected areas shall be given special considerations in their appropriation, operation and management in order to satisfy the requisites and demands for their protection, conservation, enrichment, rehabilitation, and sustenance.

Policy 2.7. Carrying capacity. Every appropriation and utilization and CCP practice-options of and for all water and related resources of the university shall be within the limits of their objectively assessed carrying capacity.

Policy 2.8. Conservation and rehabilitation. Water conservation measures like of establishment of water impoundments, catch basins, water treatment facilities, and related infrastructures shall be given importance. At the same time, activities to recharge water tables and the restoration of watersheds shall be a priority concern. Rehabilitation of "polluted" resources must be immediate.

Policy 2.9. Protection. All water and related resources of the university, ground and surface, shall be shielded from any kind of pollution, ground water exploitation, and of destruction induced by nature and/or perpetrated by man and/or animals.

Prinicple 3.0. Administration and Enforcement

Policy 3.1. University Water Resources Management (UniWaRM) Office. An office shall be created to manage and administer the water resources of the university. Where such task is an implicit or

8

Document: MMSU Water Policy, control, conservation and protection of waters (Policies 2.6–2.9), Sirmata 2040 LUDIP.

SIRMATA 2040 (2021) · INTEGRATED POLLUTION AND WASTE MANAGEMENT PROGRAM (REVISED 2020)

MMSU Water Policy Protects University Waters from Pollution

Policy · Sirmata 2040 (2021) and Integrated Pollution and Waste Management Program (revised 2020)

1 **THE BASIS**
MMSU's Water Policy under Sirmata 2040 rests on the Water Code of the Philippines and its 2005 implementing rules.

2 **POLICIES 2.9 AND 2.8**
"All water and related resources of the university, ground and surface, shall be shielded from any kind of pollution" (2.9); rehabilitation of polluted resources must be immediate, giving importance to impoundments, catch basins and water treatment facilities (2.8).

3 **WASTEWATER DISCHARGE**
The IPWMP provides for registration as a hazardous waste generator for wastewater discharge from agro-industrial facilities such as the piggery, in compliance with RA 9275, the Philippine Clean Water Act.

2021
Water Policy

RA 9275
Clean Water Act

Source: MMSU, Land Use, Zoning, Water and Energy Policies (Sirmata 2040 LUDIP, 2021) · MMSU, Revised Integrated Pollution and Waste Management Program · MMSU, Water Discharge Management (SDG 15 page)

WHERE THE STORY LANDS

The Water Policy shields university waters from pollution, the pollution programme provides for registering wastewater from agro-industrial facilities, and 17 farm wells are monitored month by month.

17	6.75–7.69	RA 9275
groundwater wells	mean pH, 2024	Clean Water Act

THE FRAMES

1. **Groundwater Quality Monitoring at the MMSU Farm**
   Feb 2024 – Jun 2025
2. **Memorandum No. 26-370 Institutionalizes MMSU's Environmental Compliance and Hazardous Waste Procedures**
   30 Sep 2026
3. **MMSU Water Policy Protects University Waters from Pollution**
   Sirmata 2040 (2021)

SOURCES (5)

1. MMSU, Groundwater Quality Assessment of the MMSU Farm (monitoring February 2024 – June 2025)
   https://www.mmsu.edu.ph/resources/15g9Xtys7q3Rujd0-CZvPH4OHTdJ1__Z8/data
2. MMSU Office of the President, Memorandum No. 26-370, "Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure," 30 September 2026
3. MMSU, Land Use, Zoning, Water and Energy Policies (Sirmata 2040 LUDIP, 2021)
   https://www.mmsu.edu.ph/resources/19uycXrgcO_iXPdPpMQwdCRxmoD1U5i60/view
4. MMSU, Revised Integrated Pollution and Waste Management Program
   https://www.mmsu.edu.ph/resources/1fNUCZwyOozOKO27L2Bm6pNN-P_cJNpLd/view
5. MMSU, Water Discharge Management (SDG 15 page)
   https://www.mmsu.edu.ph/resources/1l8foDfenNc7xpKp8wM8XShfZT58YyjSH/data